

SPEAR: Five Principles for Interactive Human-Agent Alignment

Tao Long
long@cs.columbia.edu
Columbia University
New York City, New York, USA

Lydia B. Chilton
chilton@cs.columbia.edu
Columbia University
New York City, New York, USA

Abstract

Recent AI alignment work often frames alignment as a pre-deployment optimization problem: collect human feedback, learn preferences or principles, fine-tune the model, and deploy an aligned system. This framing has produced major progress, but it under-specifies what happens once AI systems act as agents on users' behalf in situated, long-term, and social contexts. This position paper re-frames human-agent alignment as an ongoing interaction design problem. We propose **SPEAR**, five pillars of interactive alignment: Specification (how people express intent and establish shared understanding), Process (how agents decide when to act, ask, defer, or pause), Evaluation (how people judge whether agents succeeded), Adaptation (how agents adapt to users over repeated use), and Recalibration (how people adapt their trust, expectations, and behavior in response to agents).

CCS Concepts

• **Human-centered computing** → **HCI design and evaluation methods**; *HCI theory, concepts and models; concepts and paradigms.*

1 Introduction

AI alignment is often defined in machine learning as the problem of *training models to act in accordance with human intentions* [24]. This framing has produced a powerful family of pre-deployment alignment methods. Reinforcement learning from human feedback uses human preference comparisons to train reward models [6]; instruction tuning combines human demonstrations and preference data to improve helpfulness and instruction following [24]; and constitutional AI encodes explicit normative principles into model behavior [2]. Alignment is therefore commonly operationalized as a linear pipeline: collect human feedback, learn or encode an objective, fine-tune the model, and deploy an aligned system.

This model-centered framing is necessary, but increasingly insufficient for interactive AI systems, and especially for AI agents that plan, use tools, and act on users' behalf. In deployed use, human intentions are rarely complete, stable, or fully specifiable in advance. Users may not know exactly what they want until they see what the system produces. They may refine goals across turns, discover new uses, change preferences, or develop new expectations as they appropriate the system into everyday work. In this setting, alignment cannot be understood only as a property encoded into model weights before deployment. Instead, alignment emerges through interaction after deployment.

Recent HCI work makes this shift explicit. Terry et al. [29] argue that AI alignment should be understood as an ongoing, multi-stage interactive process consisting of specification alignment, process alignment, and evaluation alignment. Specification alignment concerns how humans and AI systems establish a shared understanding of the desired goal. Process alignment concerns how humans and AI systems coordinate the actions and procedures used to achieve that goal. Evaluation alignment concerns how users determine whether the AI's behavior or output actually succeeded. This framework reframes alignment from a question of output correctness alone to a question of interaction design: how should systems communicate, coordinate, and support user judgment throughout use? Chen et al. and Shen et al. extend this view by arguing that human-AI alignment is iterative and bidirectional [5, 27]. AI systems adapt to humans, but humans also adapt to AI. Users update their mental models, calibrate or miscalibrate trust, change behaviors, form social expectations, and sometimes become over-reliant on AI systems. Thus, alignment does not only concern whether AI follows human intent. It also concerns how interaction with AI reshapes human judgment, agency, and social behavior over time.

Building on these perspectives, this position paper synthesizes an **HCI definition of interactive human-AI alignment**:

Alignment is the ongoing, interactive process through which humans and AI systems develop, maintain, and repair shared understanding, appropriate agency, and calibrated reliance over time.

This definition reframes alignment as an interaction-design problem. Rather than asking only how to train a model to satisfy human preferences, it asks how humans and AI systems continuously communicate, coordinate, evaluate, adapt, and co-adapt in situated use. These questions become sharper for agents, which act rather than only respond; we therefore focus on **human-agent alignment** and ask five interactive alignment questions as **SPEAR**:

- **Specification alignment**: How do people express intent, and how do people and agents reach a shared understanding?
- **Process alignment**: How does an agent decide when to act, ask, defer, or pause in ways that match what the user would want?
- **Evaluation alignment**: How do people and agents agree on what counts as success?
- **Adaptation alignment**: What should an agent carry forward from past interactions, and what should it leave behind, so that it becomes more aligned with a user?
- **Recalibration alignment**: How do people update their trust, expectation, and behavior as they learn what agents can(not) do?

The first three pillars, **S**, **P**, and **E**, adopted from Terry et al. [29], form the core loop of interactive alignment; the last two, **A** and **R**, extend this loop over time, capturing how agents adapt to people and how people recalibrate to agents [5, 27].



We argue that these questions are not entirely new. HCI has studied closely related problems for decades under the names of grounding, activity theory, gulfs of execution and evaluation, distributed cognition, mixed-initiative interaction, adaptation, appropriation, social interaction, and groupware. What is new is the scale, opacity, agency, and adaptivity of contemporary AI systems, especially agents. This creates an opportunity to translate HCI theories and interaction mechanisms into the language of AI alignment.

2 SPEAR: Principles for Interactive Human-Agent Alignment

The five alignment questions above provide a bridge between the alignment frameworks and older foundations in HCI and related fields (as underlined). In this section, we map each question to a corresponding HCI theoretical foundation and a set of interaction mechanisms that can support alignment in deployed agents.

Specification Alignment: Intent Grounding. User goals are often partial, ambiguous, or context-dependent. Grounding and repair provide a foundation here: communication requires participants to establish common ground through clarification, acknowledgment, evidence, and repair [3, 7]. Applied to agents, this suggests that alignment requires more than instruction following. Systems need mechanisms for asking clarifying questions, surfacing assumptions, confirming interpretations, and repairing misunderstandings. Grounding further distinguishes acknowledging a request from establishing evidence of understanding [7]. For example, a user at Columbia University asks for an “affordable and convenient” flight to the Washington, DC, area for HCOMP 2026 in Alexandria, Virginia. A travel-booking agent could ask, “Would an LGA–DCA flight under \$400, departing between 9 a.m. and 6 p.m., work for you? LaGuardia is relatively close to Columbia than JFK or EWR, while DCA is closer to Alexandria than IAD.” The user can then repair this interpretation by responding, “JFK is fine if it saves at least \$100,” prompting the agent to revise its proposed airport-selection criteria and seek confirmation. Such provisional interpretations make assumptions about price, timing, and airport choice inspectable, allowing users to correct the specification before the agent commits to action.

Process Alignment: Agency Coordination. As agents act on users’ behalf, alignment failures may occur even when the final task is completed successfully. For example, a user asks, “Book my flight for the HCOMP conference.” The agent finds a suitable flight and attempts to use the company payment account, but discovers that the account has expired. It then finds a connected account on the booking page: the user’s saved personal card. Without asking for confirmation, the agent uses the personal card to complete the booking and reports only, “Your flight is booked. All set.” The task is completed, but the user remains unaware of the payment substitution. The agent has implicitly treated authorization to book the flight as authorization to choose how to pay for it. HCI work on distributed cognition and mixed-initiative interaction provides foundations for distributed cognitive work and the coordination of initiative [14–16]. AI agents allow users to offload parts of planning and decision-making, distributing cognitive work across people, models, tools, and external representations. However, offloading cognitive work does not by itself determine which decisions an

agent is authorized to make. Complementing this account, the activity theory foregrounds how tool-mediated activity is organized through purposes, rules, community, and a division of labor [11, 21]. It adds a question of legitimate delegation to the distribution of cognitive work: which decisions may an agent make, under whose authority, and with what responsibilities? Initiative is thus an alignment decision: acting, asking, deferring, or pausing should reflect not only task uncertainty but also authorization boundaries. Recent work on proactive agents and ambiguity-sensitive pausing illustrates how interaction mechanisms can support user control while enabling automation [25, 31].

Evaluation Alignment: Success Inspectability. Users evaluate agents through the evidence available to them, yet agents often obscure their capabilities, limitations, action histories, and resulting effects. Norman’s gulfs of evaluation and execution clarify why making this evidence visible must support both judgment and intervention [17]. For example, after booking the flight, the agent may report, “Your flight is confirmed.” This confirms task completion but does not establish whether the booking meets the user’s constraints. More useful feedback would state, “Your LGA–DCA flight costs \$350 and arrives at 12:30 p.m., 30 minutes later than your requested arrival deadline.” Connecting the outcome to the user’s budget and timing constraints helps bridge the gulf of evaluation. Clearly labeled options such as “Find an earlier flight” and “View cancellation terms” help bridge the gulf of execution by making corrective actions discoverable. These gulfs provide a theoretical foundation for work on HAI interaction guidelines, explainability, and agent visibility, emphasizing the need for helping users understand system capabilities, ongoing actions, and opportunities for correction [1, 4, 30]. Visibility into agent state, action traces, goal representations, and side effects is therefore an alignment mechanism when it enables users to compare outcomes with their intentions and respond to mismatches, rather than merely exposing more information.

Adaptation Alignment: Over Time. Alignment must remain revisable as users change their goals and incorporate agents into everyday work. HCI work on adaptation and appropriation shows how users repurpose technologies and develop new practices through use [9, 10, 20]. Structuration theory describes rules and resources as both shaping action and being reproduced or transformed through it [12, 28]. Orlikowski’s practice lens applies this relationship to recurring technology use [23]. For example, while planning travel to HCOMP 2026, a user says, “JFK is fine if it saves at least \$100.” If saved and repeatedly accepted, this situational trade-off may become a default for future trips, even when shorter ground travel becomes more important. Similarly, when a company payment account is unavailable, the user may authorize a temporary workaround: “Use my personal card this time; I will request reimbursement.” If the agent subsequently proposes that card and the user repeatedly accepts it for convenience, the workaround may become routine, leaving the user responsible for advancing funds and requesting reimbursement. In both cases, system defaults shape subsequent actions, while repeated acceptance reinforces those defaults without necessarily establishing their continued appropriateness. This lens motivates editable alignment artifacts—preferences, policies, permissions, and histories—and opportunities to reconsider their

persistence: “*Keep the \$100 airport trade-off for future trips?*” or “*The company account is available again. Switch back?*” Longitudinal evaluation should examine how such arrangements become routine, how users appropriate agents into different roles, and whether earlier choices remain open to revision [19, 26].

Recalibration Alignment: Human-to-Agent Alignment. Alignment does not only shape agent behavior; interaction also changes how people understand, use, and rely on agents. This bidirectional view treats human adaptation not as alignment in itself, but as a process through which people develop more appropriate ways of interpreting and collaborating with agents. Works on computers as social actors, groupware and social interaction highlights how people respond socially to computers and adapt their coordination practices to technology-mediated work [13, 22]. Dennett’s *intentional stance* describes predicting a system’s behavior by attributing beliefs and desires to it, without requiring that it possess human-like mental states [8]. For example, after several successful travel bookings, a user may reason, “*It knows my preferences and wants to find what works best for me.*” Such interactions can reshape:

- expectations (“*It should check baggage fees without being asked*”),
- trust (“*Its previous recommendations were good, so I trust this*”),
- mental models (“*It will now remember my \$100 airport trade-off*”),
- practices (“*I now check only the summary, not the full itinerary*”),
- delegation boundaries (“*Book flights under \$400 without asking me each time*”).

These adaptations do not necessarily constitute improved alignment. Alignment concerns whether users’ evolving expectations, mental models, and reliance become *calibrated to the system’s actual capabilities, behavior, and limitations*. A user who comes to expect persistent memory when the system cannot reliably retain preferences has adapted to the system, but in a way that creates a mismatch between their mental model and the agent. Capability demonstrations and expectation-setting mechanisms [18], including explicit memory boundaries such as “*Your airport preference applies to this booking only; it has not been saved for future trips,*” can help users calibrate their interpretations and reliance. Human-to-agent alignment should therefore support *informed and revisable calibration*, rather than assume that greater trust, less checking, or broader delegation is necessarily beneficial.

3 Summary: Interaction Mechanisms as Alignment Mechanisms

Across these five questions, we argue that interaction design mechanisms can be understood as alignment mechanisms. Clarification, pausing, visibility, onboarding, correction, explanation, customization, and expectation-setting are not merely interface details; they are the means by which alignment is negotiated, maintained, and repaired after deployment.

We see our work contributes a bridge between AI alignment, human computation, and human-AI interaction. Human feedback is central to modern alignment pipelines, but the role of humans is often compressed into labels, rankings, demonstrations, or evaluations. An HCI perspective expands the role of humans from

feedback providers to situated collaborators, evaluators, appropriators, and co-adapting participants in sociotechnical systems. This shift is especially important for agentic systems that act over time, coordinate across people and tools, and shape collective work.

References

- [1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI ’19). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3290605.3300233
- [2] Yuntao Bai, Saurav Kadavath, Sandipan Kundu, Amanda Askell, Jackson Kernion, Andy Jones, Anna Chen, Anna Goldie, Azalia Mirhoseini, Cameron McKinnon, Carol Chen, Catherine Olsson, Christopher Olah, Danny Hernandez, Dawn Drain, Deep Ganguli, Dustin Li, Eli Tran-Johnson, Ethan Perez, Jamie Kerr, Jared Mueller, Jeffrey Ladish, Joshua Landau, Kamal Ndousse, Kamile Lukosuite, Liane Lovitt, Michael Sellitto, Nelson Elhage, Nicholas Schiefer, Noemi Mercado, Nova DasSarma, Robert Lasenby, Robin Larson, Sam Ringer, Scott Johnston, Shauna Kravec, Sheer El Showk, Stanislav Fort, Tamera Lanham, Timothy Telleen-Lawton, Tom Conerly, Tom Henighan, Tristan Hume, Samuel R. Bowman, Zac Hatfield-Dodds, Ben Mann, Dario Amodei, Nicholas Joseph, Sam McCandlish, Tom Brown, and Jared Kaplan. 2022. Constitutional AI: Harmlessness from AI Feedback. arXiv:2212.08073 [cs.CL]. <https://arxiv.org/abs/2212.08073>
- [3] Susan E. Brennan. 1998. The Grounding Problem in Conversations with and through Computers. In *Social and Cognitive Approaches to Interpersonal Communication*, Susan R. Fussell and Roger J. Kreuz (Eds.). Lawrence Erlbaum Associates, 201–225.
- [4] Alan Chan, Carson Ezell, Max Kaufmann, Kevin Wei, Lewis Hammond, Herbie Bradley, Emma Bluemke, Nitarshan Rajkumar, David Krueger, Noam Kolt, Lennart Heim, and Markus Anderljung. 2024. Visibility into AI Agents. In *Proceedings of the 2024 ACM Conference on Fairness, Accountability, and Transparency* (Rio de Janeiro, Brazil) (FAccT ’24). Association for Computing Machinery, New York, NY, USA, 958–973. doi:10.1145/3630106.3658948
- [5] Valerie Chen, Cleotilde Gonzalez, Anita Williams Woolley, Michael Lee, Tongshuang Wu, Vincent Conitzer, and Aarti Singh. 2026. Align AI to Dynamic Human-AI Workflows. arXiv:2607.14240 [cs.AI] version: 1. doi:10.48550/arXiv.2607.14240
- [6] Paul F. Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. 2017. Deep reinforcement learning from human preferences. In *Proceedings of the 31st International Conference on Neural Information Processing Systems* (Long Beach, California, USA) (NIPS’17). Curran Associates Inc., Red Hook, NY, USA, 4302–4310.
- [7] Herbert H. Clark and Susan E. Brennan. 1991. Grounding in Communication. In *Perspectives on Socially Shared Cognition*, Lauren B. Resnick, John M. Levine, and Stephanie D. Teasley (Eds.). American Psychological Association, Washington, DC, 127–149. doi:10.1037/10096-006
- [8] Daniel C. Dennett. 1989. *The intentional stance*. MIT press.
- [9] Alan Dix. 2007. Designing for appropriation. In *Proceedings of the 21st British HCI Group Annual Conference on People and Computers: HCI...but Not as We Know It - Volume 2* (University of Lancaster, United Kingdom) (BCS-HCI ’07). BCS Learning & Development Ltd., Swindon, GBR, 27–30.
- [10] Paul Dourish. 2003. The Appropriation of Interactive Technologies: Some Lessons from Placeless Documents. *Comput. Supported Coop. Work* 12, 4 (Sept. 2003), 465–490. doi:10.1023/A:1026149119426
- [11] Yrjö Engeström. 2000. Activity theory as a framework for analyzing and redesigning work. *Ergonomics* 43, 7 (2000), 960–974.
- [12] Anthony Giddens. 2014. Structuration theory: past, present and future. In *Giddens’ theory of structuration*. Routledge, 201–221.
- [13] Jonathan Grudin. 1994. Groupware and social dynamics: eight challenges for developers. *Commun. ACM* 37, 1 (Jan. 1994), 92–105. doi:10.1145/175222.175230
- [14] James Hollan, Edwin Hutchins, and David Kirsh. 2000. Distributed cognition: toward a new foundation for human-computer interaction research. *ACM Trans. Comput.-Hum. Interact.* 7, 2 (June 2000), 174–196. doi:10.1145/353485.353487
- [15] Eric Horvitz. 1999. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Pittsburgh, Pennsylvania, USA) (CHI ’99). Association for Computing Machinery, New York, NY, USA, 159–166. doi:10.1145/302979.303030
- [16] Edwin Hutchins and Tove Klausen. 1996. Distributed Cognition in an Airline Cockpit. In *Cognition and Communication at Work*, Yrjö Engeström and David Middleton (Eds.). Cambridge University Press, Cambridge, 15–34.
- [17] Edwin L. Hutchins, James D. Hollan, and Donald A. Norman. 1985. Direct manipulation interfaces. *Human-computer interaction* 1, 4 (1985), 311–338.
- [18] Rafal Kocielnik, Saleema Amershi, and Paul N. Bennett. 2019. Will You Accept an Imperfect AI? Exploring Designs for Adjusting End-user Expectations of AI Systems. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing*

- Systems* (Glasgow, Scotland Uk) (*CHI '19*). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3290605.3300641
- [19] Tao Long, Katy Ilonka Gero, and Lydia B Chilton. 2024. Not Just Novelty: A Longitudinal Study on Utility and Customization of an AI Workflow. In *Proceedings of the 2024 ACM Designing Interactive Systems Conference* (Copenhagen, Denmark) (*DIS '24*). Association for Computing Machinery, New York, NY, USA, 782–803. doi:10.1145/3643834.3661587
- [20] Thomas P Moran. 2002. Everyday adaptive design. In *Proceedings of the 4th Conference on Designing Interactive Systems: Processes, Practices, Methods, and Techniques* (London, England) (*DIS '02*). Association for Computing Machinery, New York, NY, USA, 13–14. doi:10.1145/778712.778715
- [21] Bonnie A Nardi. 1996. Activity theory and human-computer interaction. *Context and consciousness: Activity theory and human-computer interaction* 436, 1 (1996), 7–16.
- [22] Clifford Nass, Jonathan Steuer, and Ellen R. Tauber. 1994. Computers are social actors. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems* (Boston, Massachusetts, USA) (*CHI '94*). Association for Computing Machinery, New York, NY, USA, 72–78. doi:10.1145/191666.191703
- [23] Wanda J Orlikowski. 2000. Using technology and constituting structures: A practice lens for studying technology in organizations. *Organization science* 11, 4 (2000), 404–428.
- [24] Long Ouyang, Jeff Wu, Xu Jiang, Diogo Almeida, Carroll L. Wainwright, Pamela Mishkin, Chong Zhang, Sandhini Agarwal, Katarina Slama, Alex Ray, John Schulman, Jacob Hilton, Fraser Kelton, Luke Miller, Maddie Simens, Amanda Askell, Peter Welinder, Paul Christiano, Jan Leike, and Ryan Lowe. 2022. Training language models to follow instructions with human feedback. In *Proceedings of the 36th International Conference on Neural Information Processing Systems* (New Orleans, LA, USA) (*NIPS '22*). Curran Associates Inc., Red Hook, NY, USA, Article 2011, 15 pages.
- [25] Yi-Hao Peng, Dingzeyu Li, Jeffrey P Bigham, and Amy Pavel. 2025. Morae: Proactively Pausing UI Agents for User Choices. In *Proceedings of the 38th Annual ACM Symposium on User Interface Software and Technology* (*UIST '25*). Association for Computing Machinery, New York, NY, USA, Article 198, 14 pages. doi:10.1145/3746059.3747797
- [26] Steven I. Ross, Fernando Martinez, Stephanie Houde, Michael Muller, and Justin D. Weisz. 2023. The Programmer's Assistant: Conversational Interaction with a Large Language Model for Software Development. In *Proceedings of the 28th International Conference on Intelligent User Interfaces* (Sydney, NSW, Australia) (*IUI '23*). Association for Computing Machinery, New York, NY, USA, 491–514. doi:10.1145/3581641.3584037
- [27] Hua Shen, Tiffany Kneare, Reshmi Ghosh, Michael Xieyang Liu, Andrés Monroy-Hernández, Tongshuang Wu, Diyi Yang, Yun Huang, Tanushree Mitra, Yang Li, and Marti Hearst. 2025. Bidirectional Human-AI Alignment: Emerging Challenges and Opportunities. In *Proceedings of the Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* (*CHI EA '25*). Association for Computing Machinery, New York, NY, USA, Article 857, 6 pages. doi:10.1145/3706599.3716291
- [28] Rob Stones. 2017. *Structuration theory*. Bloomsbury Publishing.
- [29] Michael Terry, Chinmay Kulkarni, Martin Wattenberg, Lucas Dixon, and Meredith Ringel Morris. 2023. Interactive AI Alignment: Specification, Process, and Evaluation Alignment. arXiv:2311.00710
- [30] Qian Yang, Aaron Steinfeld, Carolyn Rosé, and John Zimmerman. 2020. Re-examining Whether, Why, and How Human-AI Interaction Is Uniquely Difficult to Design. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems* (Honolulu, HI, USA) (*CHI '20*). Association for Computing Machinery, New York, NY, USA, 1–13. doi:10.1145/3313831.3376301
- [31] Nima Zargham, Leon Reicherts, Michael Bonfert, Sarah Theres Voelkel, Johannes Schoening, Rainer Malaka, and Yvonne Rogers. 2022. Understanding Circumstances for Desirable Proactive Behaviour of Voice Assistants: The Proactivity Dilemma. In *Proceedings of the 4th Conference on Conversational User Interfaces* (Glasgow, United Kingdom) (*CUI '22*). Association for Computing Machinery, New York, NY, USA, Article 3, 14 pages. doi:10.1145/3543829.3543834